

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

A data analytics project using Python, PostgreSQL, and Power BI

Dataset	3,900 purchases
Columns	18
Primary Tools	Python • PostgreSQL • Power BI
Focus Areas	Customer segments • Spending • Products • Subscriptions

Project objective: uncover actionable insights into spending patterns, customer segments, product preferences, and subscription behavior to support business decisions.

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

Attribute	Summary
Rows	3,900
Columns	18
Customer demographics	Age, Gender, Location, Subscription Status
Purchase details	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, S
Missing data	37 values in Review Rating

Key dataset dimensions

The analysis combines customer demographics, purchase attributes, shopping behavior, reviews, discounts, subscriptions, and shipping information. These fields support both customer-level segmentation and category-level business analysis.

3. Exploratory Data Analysis using Python

Data preparation and cleaning

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.
- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:** Created `age_group` by binning customer ages and created `purchase_frequency_days` from purchase data.
- **Data Consistency Check:** Verified whether `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected the Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

Analysis workflow

Step	Purpose
Load	Bring transactional data into pandas
Explore	Understand structure and summary statistics
Clean	Handle missing values and standardize columns
Engineer	Create age groups and purchase-frequency features
Validate	Check consistency and redundancy
Integrate	Load cleaned data into PostgreSQL

4. Data Analysis using SQL

Structured analysis was performed in PostgreSQL to answer business questions around revenue, discounts, products, shipping, subscriptions, customer segments, and age groups.

- 1. Revenue by Gender** — Compared total revenue generated by male vs. female customers.
- 2. High-Spending Discount Users** — Identified customers who used discounts but still spent above the average purchase amount.
- 3. Top 5 Products by Rating** — Found products with the highest average review ratings.
- 4. Shipping Type Comparison** — Compared average purchase amounts between Standard and Express shipping.
- 5. Subscribers vs. Non-Subscribers** — Compared average spend and total revenue across subscription status.
- 6. Discount-Dependent Products** — Identified 5 products with the highest percentage of discounted purchases.
- 7. Customer Segmentation** — Classified customers into New, Returning, and Loyal segments based on purchase history.
- 8. Top 3 Products per Category** — Listed the most purchased products within each category.
- 9. Repeat Buyers & Subscriptions** — Checked whether customers with more than 5 purchases are more likely to subscribe.
- 10. Revenue by Age Group** — Calculated total revenue contribution of each age group.

5. Power BI Dashboard

The final dashboard presents the customer shopping analysis visually and interactively. It includes KPI cards, category analysis, subscription status, and age-group comparisons.



Dashboard components visible in the supplied design: subscription status and gender filters; category and shipping filters; KPI cards for average purchase amount, average review rating, and customer count; sales and revenue by category; subscription-status distribution; and sales/revenue by age group.

6. Business Recommendations

1. **Boost Subscriptions** — Promote exclusive benefits for subscribers.
2. **Customer Loyalty Programs** — Reward repeat buyers to move them into the “Loyal” segment.
3. **Review Discount Policy** — Balance sales boosts with margin control.
4. **Product Positioning** — Highlight top-rated and best-selling products in campaigns.
5. **Targeted Marketing** — Focus efforts on high-revenue age groups and express-shipping users.

Business takeaway

The project combines data cleaning, feature engineering, SQL-based business analysis, and Power BI visualization into one end-to-end analytics workflow. The resulting analysis can be used to understand customer behavior and prioritize subscription, loyalty, product, discount, and marketing initiatives.

7. End-to-End Project Summary

Stage	Technology	Outcome
Data preparation	Python / pandas	Clean and standardized dataset
Feature engineering	Python	Age groups and purchase-frequency features
Database analysis	PostgreSQL / SQL	Business questions answered with transactions
Visualization	Power BI	Interactive customer behavior dashboard
Decision support	Analytics	Recommendations for subscriptions, loyalty, products, discounts, and marketing

This report is based on the supplied Customer Shopping Behavior Analysis document and the supplied Power BI dashboard image. It preserves the project's stated dataset, workflow, analysis questions, and recommendations without adding unsupported business findings.